

Final Project Summary

EMS Readiness Optimization for Manhattan — v1.1.0

Completion Date: March 12, 2026
Repository: github.com/cnsp/ems-optimization
Tag: v1.1.0 (Gap Closure Release)

Repository Statistics

Metric	Value
Total files in repository	205
Total Python lines of code	7,348
Python source files	42
Jupyter notebooks	8
CSV result files	42
PNG figures	47
PDF documents	21
Markdown documents	26
LaTeX tables	4
YAML configuration files	5
Unit tests	39 (all passing)

Simulation Statistics

Metric	Value
Total simulation runs	1,770
Replications per cell	30
Simulation duration per run	168 hours (1 week)
Warm-up period per run	24 hours
Total simulated time	296,520 hours (~33.8 years)
Total experiments	5 (factorial design + CBD robustness)
Verification tests	4 (all passed)
Validation pilots	3 (all passed)

Key Metrics Achieved

Metric	P0 (Baseline)	P2 (Optimized)	Improvement
Mean Response Time	8.08 min	2.57 min	−68.2%
P90 Response Time	19.47 min	3.76 min	−80.7%
8-min Coverage	64.4%	99.6%	+35.2 pp
Mean Utilization	9.1%	7.5%	−1.6 pp

Statistical Significance: All results $p < 0.001$, Cohen’s $d > 28$ (P0 vs P2)

Phase Completion Status

Phase	Description	Status	Deliverables
Phase 1	Data Processing & EDA	✓ Complete	10 figures, processed data, EDA notebook
Phase 2	Demand & Service Modeling	✓ Complete	NHPP model, lambda tables, 2 spec docs
Phase 3	Optimization Models	✓ Complete	3 MIP models, allocations, comparison
Phase 4	DES Simulation + V&V	✓ Complete	SimPy engine, 39 tests, V&V log
Phase 5	Production Experiments	✓ Complete	1,770 runs, 5 experiments (incl. CBD)
Phase 6	Statistical Analysis	✓ Complete	ANOVA, post-hoc, 5 pub figures, 4 tables
Phase 7	Final Report & Docs	✓ Complete	Technical report, presentation, roadmap

All 7 phases complete. Project delivered.

Files Added in This Session

Part 1: Repository Sync

- Force-added 38 files previously excluded by .gitignore
- Includes all processed data, figures, experiment tables, and maps
- Pushed successfully to origin/main

Part 2: Phase 7 Deliverables

- docs/technical_report.md (.pdf) — 25-page comprehensive report
- docs/executive_presentation.md (.pdf) — 10-slide stakeholder deck
- docs/implementation_roadmap.md (.pdf) — 3-phase deployment plan
- docs/project_archive.md (.pdf) — Timeline, decisions, lessons learned
- docs/code_documentation.md (.pdf) — Architecture & API guide
- docs/file_inventory.md (.pdf) — Complete file listing
- README.md — Enhanced with key findings and full documentation
- scripts/generate_summary_dashboard.py — Dashboard generator
- results/figures/project_summary_dashboard.png — Visual summary
- docs/final_summary.md — This file

Part 3: Gap Closure (v1.1.0)

- `docs/cbd_definition.md` — CBD geographic definition (10 precincts)
- `configs/cbd_scenario.yaml` — CBD experiment configuration
- `scripts/run_cbd_experiment.py` — CBD DES experiment runner (330 runs)
- `scripts/analyze_queue_metrics.py` — Queue metrics analysis script
- `scripts/analyze_seasonal_patterns.py` — Seasonal pattern analysis script
- `notebooks/09_cbd_analysis.ipynb` — CBD analysis notebook
- `docs/cbd_robustness_analysis.md` (.pdf) — CBD findings document
- `docs/queue_analysis.md` (.pdf) — Queue metrics analysis document
- `docs/gap_closure_report.md` (.pdf) — Gap closure verification report
- `results/simulation/cbd_experiment/` — 330 CBD simulation runs
- `results/figures/cbd_*.png` — 3 CBD figures
- `results/figures/queue_*.png` — 4 queue analysis figures
- `results/figures/seasonal_*.png` — 3 seasonal analysis figures
- `results/tables/cbd_*.csv` — 2 CBD tables
- `results/tables/queue_*.csv` — 2 queue tables
- `results/tables/seasonal_*.csv` — 1 seasonal table

v1.2.0 - Alternative Analyses (March 12, 2026)

New experiments:

- **Distance Metric Comparison:** Haversine vs Manhattan (taxicab) distance — confirmed model robustness; allocations differ at only 2/48 firehouses; simulation performance identical
- **CBD-Focused Optimisation:** Manhattan-wide vs CBD-focused allocation — demonstrated that demand-weighted objective already optimally balances CBD and non-CBD performance; CBD-focused allocation provides no CBD benefit while severely degrading non-CBD service

New deliverables:

- `scripts/generate_manhattan_distance_matrix.py` — Manhattan distance matrix generation
- `scripts/run_distance_comparison_experiment.py` — Distance metric comparison experiment
- `scripts/run_cbd_focused_optimization.py` — CBD-focused optimisation experiment
- `data/processed/distance_matrix_firehouse_precinct_manhattan.csv` — Manhattan distance matrix
- `docs/distance_metric_comparison.md` (.pdf) — Distance metric comparison report
- `docs/cbd_focused_optimization_analysis.md` (.pdf) — CBD-focused analysis report
- `docs/alternative_analyses_summary.md` (.pdf) — Combined summary of alternative analyses
- `results/distance_comparison/` — 4 figures + 2 tables
- `results/cbd_focused_comparison/` — 3 figures + 2 tables

Modified files:

- `src/ems_readiness/utils/distance.py` — Added `manhattan_distance()` function
- `src/ems_readiness/optimization/models.py` — Added `build_cbd_focused_demand_weighted()` and `build_cbd_focused_coverage()`
- `docs/technical_report.md` — Added §5.10 and §5.11; updated figures/tables lists

Project complete. Tagged as v1.2.0 — 100% alignment with project outline plus alternative analyses.